

Openvoxels Summer 2024

Uploading the most complete snapshot of the world to arweave, wearables art contest results, VRM avatar updates, Voxels Passports, and more.

June

Smooth Voxels 2.3.0 was released, offering full threejs compatibility support and some new markdown features. Smooth Voxels is powerful markdown for voxel project that we used it in the past to convert vox models to glTF for wearables. Have yet to update the smoothvoxels-tools CLI tools to the latest version, good first issue for anyone looking to contribute.

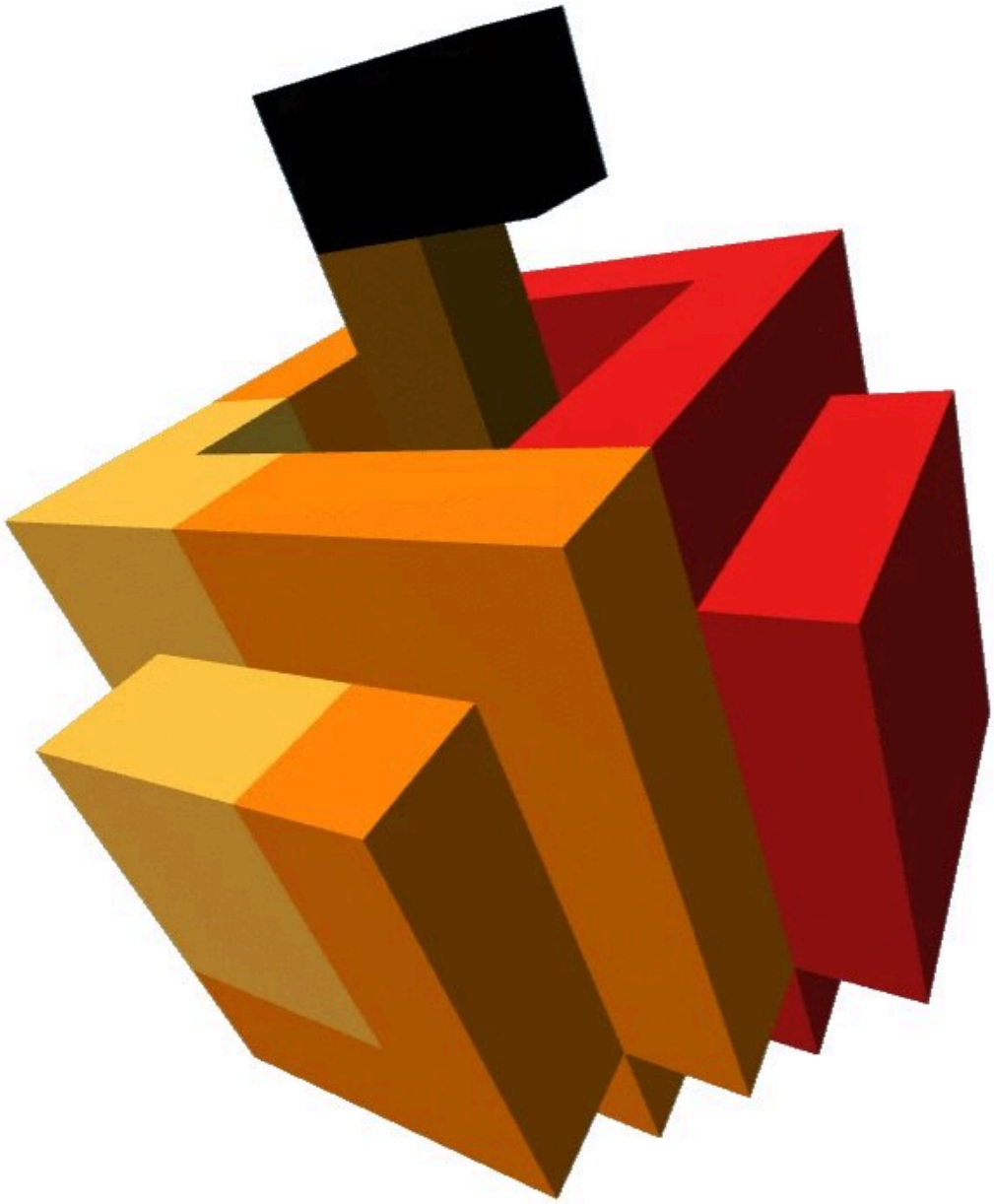
Openvoxels.eth

@openvoxels

Smooth Voxels 2.3.0 was released! Some new features include:
- Full #threejs support in playground / as library!

- New markdown (bend, smooth scale, rotate, translate models)
- Playground: New examples + Save as threejs + animation + @aframevr options

smoothvoxels.glitch.me



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[7:14 PM • Jun 3, 2024]

Starting to see some voxel creators publish to Mona’s platform which recently launched a marketplace dedicated to 3D art. It’s great to see enthusiasm and emphasis on files. Openvoxels has collected a few of such works.



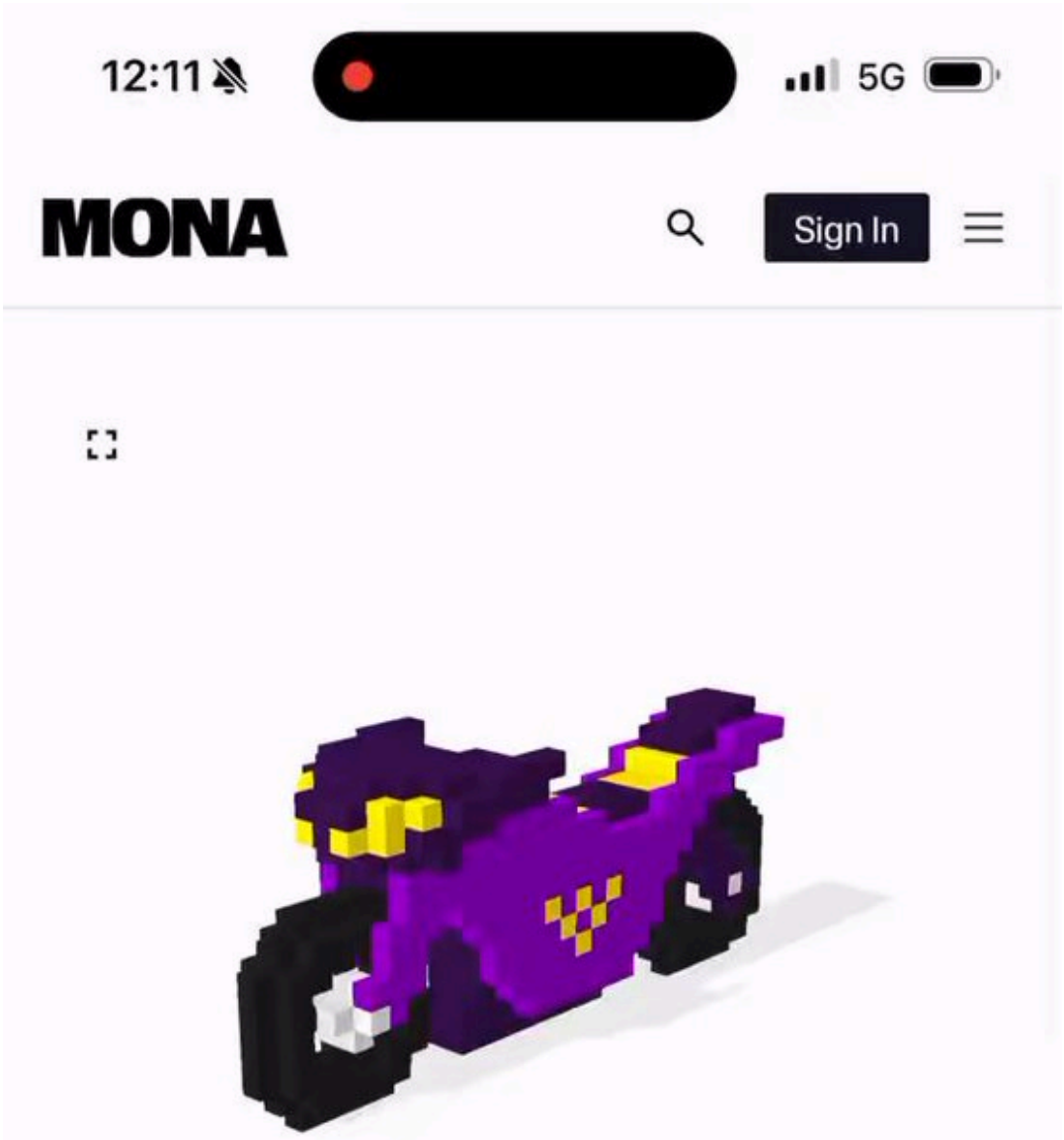
Justin Melillo ᔓ MONA

@justinmmelillo

gm from this super slick stunt motorbike by @___Philosopher_ curated for this week’s @monaverse claim!

Free on Base (link in comments below)

🏍️🔥

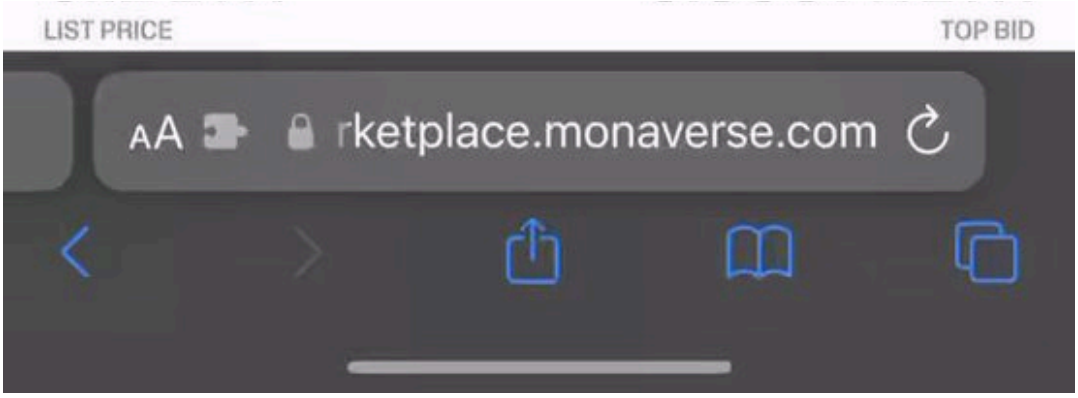


MONA Claim

Crossworlds-T3 by The Philosopher

x42

0.12 ETH 0.0001 WETH



11

3:22 PM • Jun 16, 2024

Broken Links Everywhere

Midway through June citizens of Voxels were hit by a bit of a shocking moment to discover broken links everywhere. It highlighted a point of failure due to file storage centralization. This day inspired jin to upload all vox models to arweave for posterity.



source: https://x.com/dankvr/status/1802741370936820168

Additional details

- There were 42341 unique vox models / ~1.6 GB in the snapshot
- All parcel media links are very prone to link-rot: only 12144 of 205631 were hosted on ipfs/arweave (just 5%)
    - Even IPFS has link rot issues as I've seen first hand when archiving Alotta Money's builds cuz he used a pinning service that shut down (eternum) ☹
- Picked arweave to make a backup of all vox models to for posterity, images would of been too expensive at the moment
    - There's ~75k unique images in-world, 100gb
    - ~75 AR or roughly $2000 USD, not bad for a yearly archive...
- It was just $31 in fees to upload all the vox models (https://ar-fees.arweave.dev)

Wearables Art Contest

After previously converting all the wearables to svox and glTF 2.0 we launched an art contest for people to get hands on with the files and showcase interoperability. One major aspect was to think outside the box, use wearables in NEW ways: add them to other avatars, create Blender renders, import them into other game engines, make worlds with them, etc.



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Wearables Art Contest

Showcase interoperability with new wearables glTF files with something unique, win prizes. Think beyond @cryptovoxels: blender renders, custom avatars etc!

Deadline: June 30th, submit here: jokerace.io/contest/polygo...

More info: hackmd.io/@XR/wearables-...



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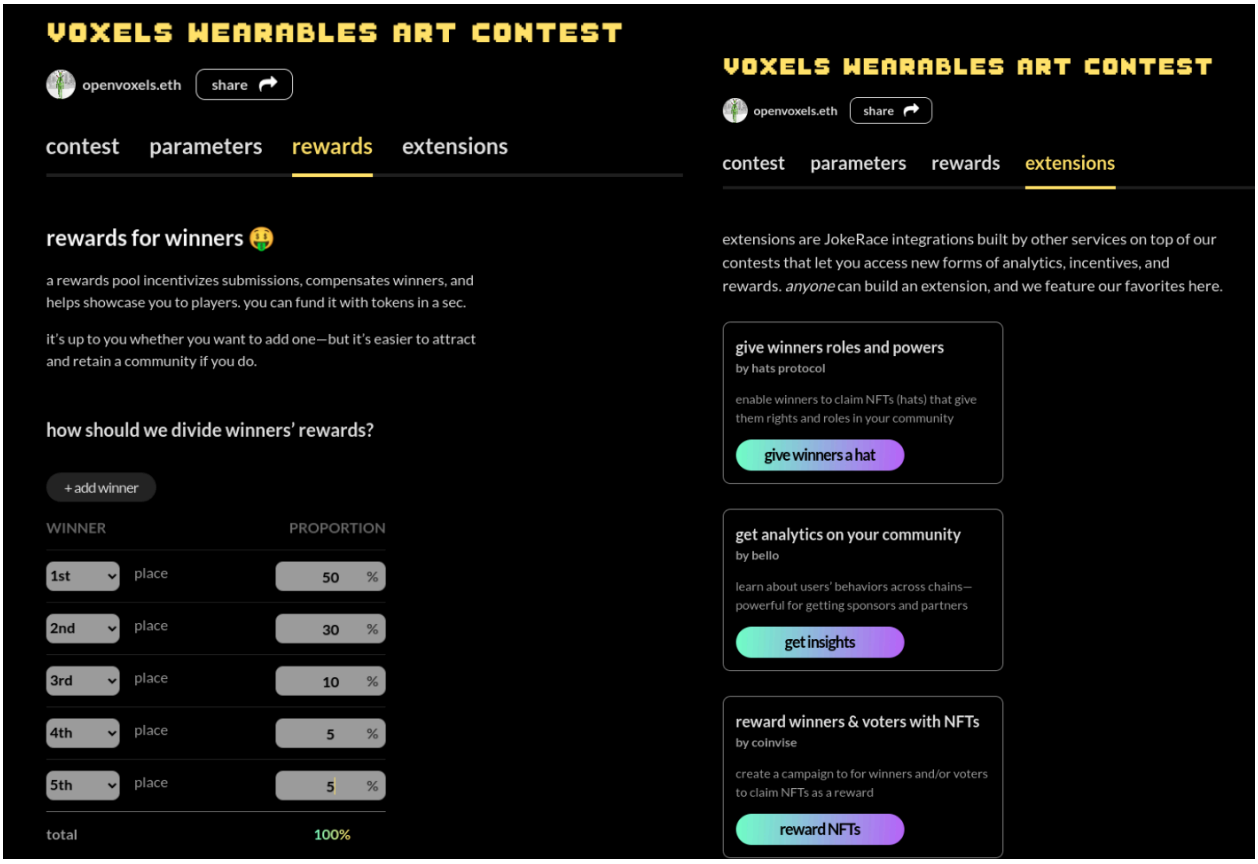
6:31 PM • Jun 16, 2024

Often the best way to learn things in crypto is to do things. The contest was also an opportunity for us to test pilot Jokerace, a platform that makes it easy to launch programmable contests on-chain, and for us to field test our latest voting power recipe.

https://www.jokerace.io/contest/polygon/0x80aa9bda9fb9e3fe262a2466bd4fee8086a3dec3

Voting power was calculated with a combination of data from parcel owners, wearables creators (based on how many you made, how many unique, how many ppl own), and Openvoxels supporters. The community voted for the winners competing for $1000 MATIC prize pool. Everyone who participated and voted also got a small gift (Openvoxels pin) as a token of appreciation.

There were some interesting extestions that Jokerace offers which could be useful if we want to explore things like using Jokerace to elect people into governance positions or distribute automatic NFT rewards that people can claim for winning / voting.



Overall Jokerace was pretty smooth, would recommend / use again

Here's a look at the winning contest entries. One was even interactive, see the claw machine made by Anarchy Arcade! https://smsithlord.com/openvoxels/clawgame



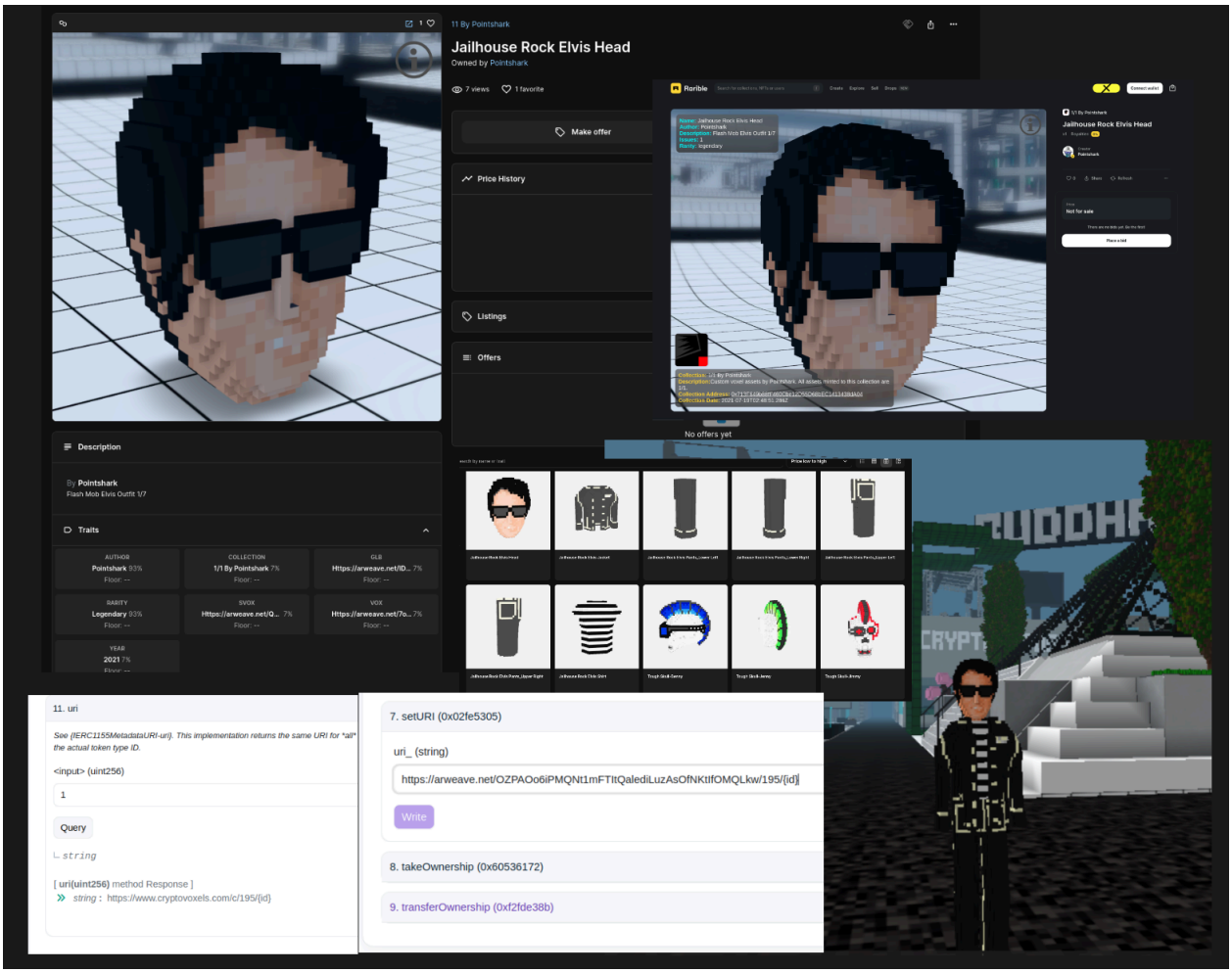
Top 5 winners from the Voxels Wearables Art Contest

July

Updated Wearables Collection with New Metadata

We upgraded an existing wearables collection through polygonscan UI with new metadata, glTF files, previews (including AR/VR), and it didn't break http://voxels.com compatibility. Thanks for being brave pioneer Pointshark!

- https://www.voxels.com/collections/195
- https://opensea.io/assets/matic/0x713F649b88fF460Cbe12D55D68bEC1413438dA04/1
- https://rarible.com/token/polygon/0x713f649b88ff460cbe12d55d68bec1413438da04:1?tab=overview

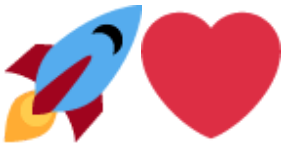


The previews updated on Opensea / Rarible, everything still worked in Cryptovoxels. SUCCESS!

We have been also chatting with Justin from Monaverse and they were able to integrate the new wearables to their marketplace too. More marketplaces, more visibility and reach, better value for the creator and platforms to expand influence.

[MONA]

MONA [@monaverse]

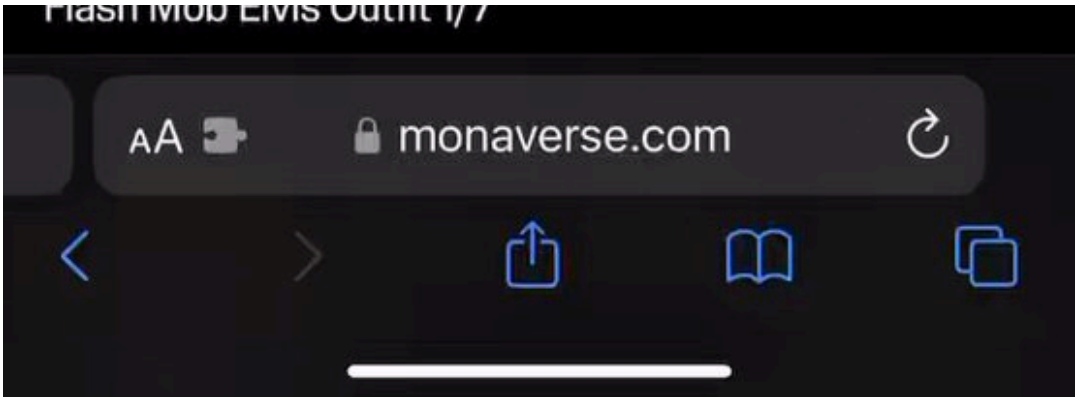


It works so well! @openvoxels + @cryptovoxels + @monaverse =



Feat. Work by the incredible @Pointshark1


1:55
5G
MONA
Sign In
11 By Pointshark
Jailhouse Rock Elvis Head
Make Offer




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Upgrading the smart contract with the new wearables metadata has made it much easier for third party marketplaces like @monaverse to integrate:
monaverse.com/collections/13...

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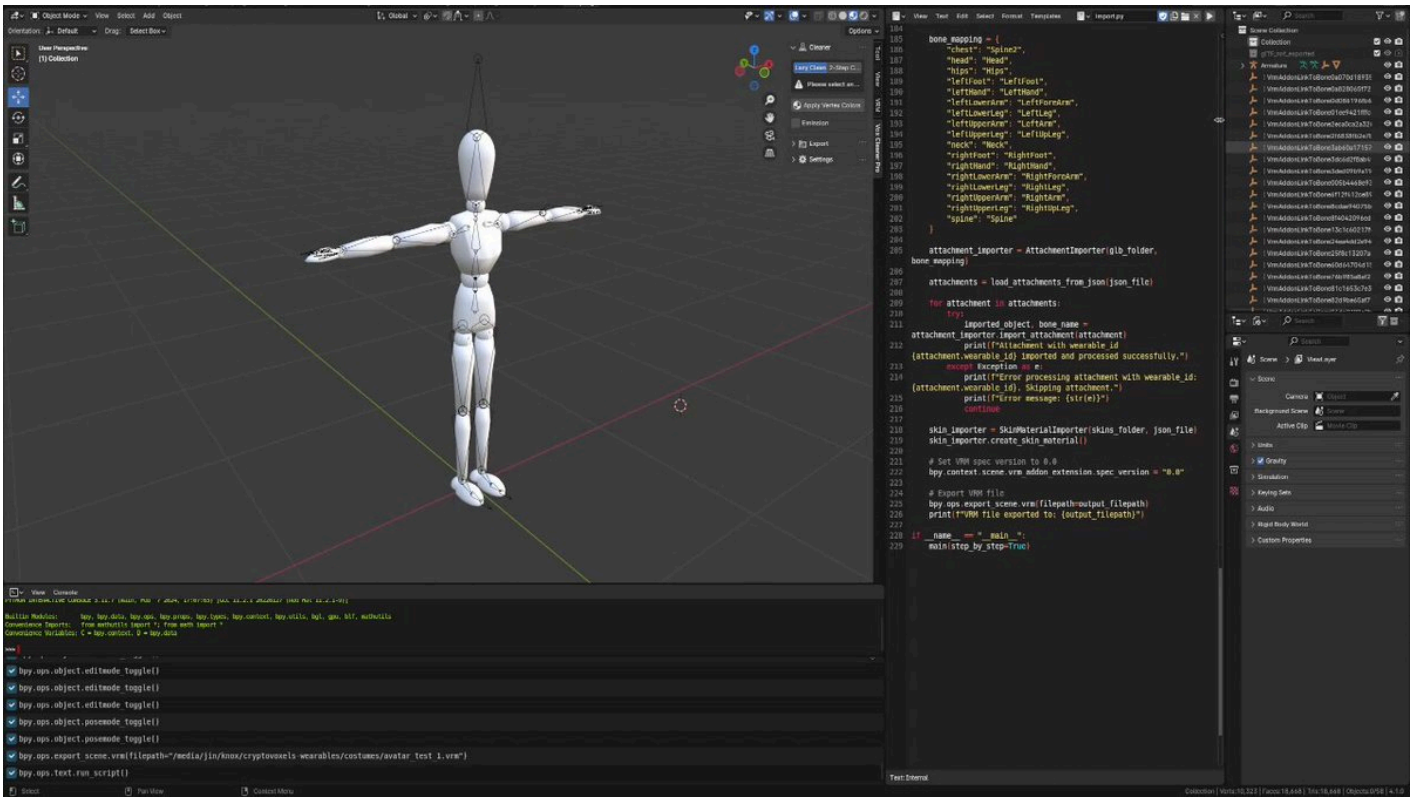
7:57 AM • Jul 28, 2024

VRM Assembly Improvements

Made the wearables → VRM avatar pipeline more accurate than before. The step by step blender pipeline offers those working on digital fashion wearables for avatar interop a look at each step for converting between different coordinate systems and rotation orders of Babylonjs and Blender.


jin
@dankvr

Learned how to visualize blender scripts step by step, useful for debugging! Here's a vid of @cryptovoxels VRM assembly (avatar costume by @ku614)



78

4:38 PM • Jul 21, 2024

News: Overture Maps Foundation Released Public Dataset

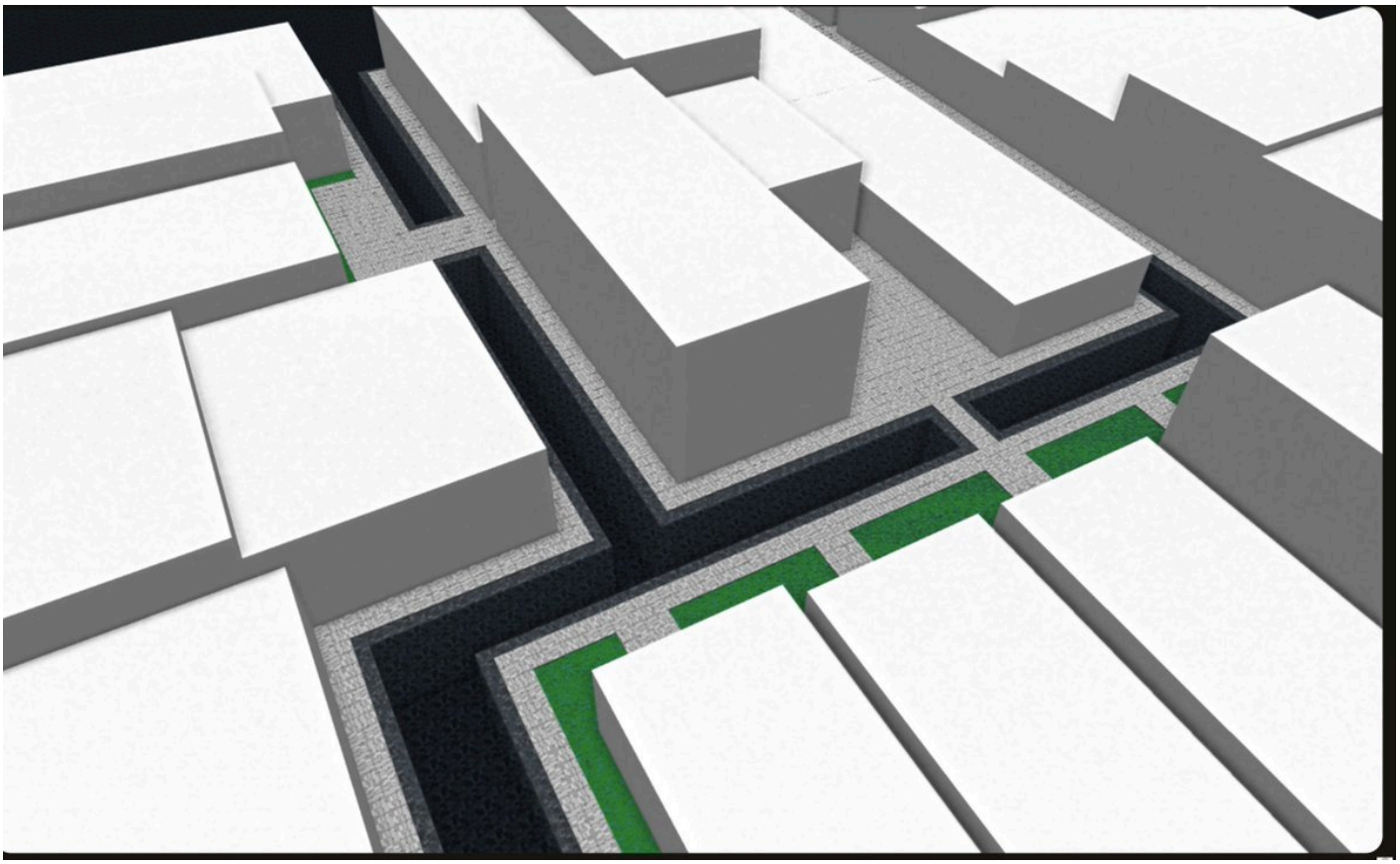
Huge public dataset release by folks from Microsoft, Meta, and Amazon working on an open-source mapping project as an alternative to mapping data provided by Google/Apple. Devs can access 2.3B unique buildings, 54MM POIs (point of interest), 200MM addresses, and more. Downloads are JSON that describes the shape of the data as well as features, geometries, properties, etc. Even without the satellite textures of Google/Apple products it still packs a ton of useful information..
https://explore.overturemaps.org/#15/38.90678/-77.03649



jin

@dankvr

Curious by the similarities and differences from virtual land platforms like @cryptovoxels. I wonder if lessons learned from experimenting with fantasy land can transcend into real world use cases / network states



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1:27 PM • Jul 25, 2024

In a way I see these blockchain based virtual worlds sort of like a fantasy simulation for independent researchers exploring digital property ownership via smart contracts vs traditional legal structures / title deeds. All feels relevant to building The Network State.

August

> Voxels team is now just Ben:
> https://x.com/cryptovoxels/status/1819138618780536908

Fixed Issue for Voxels VRM Sprites

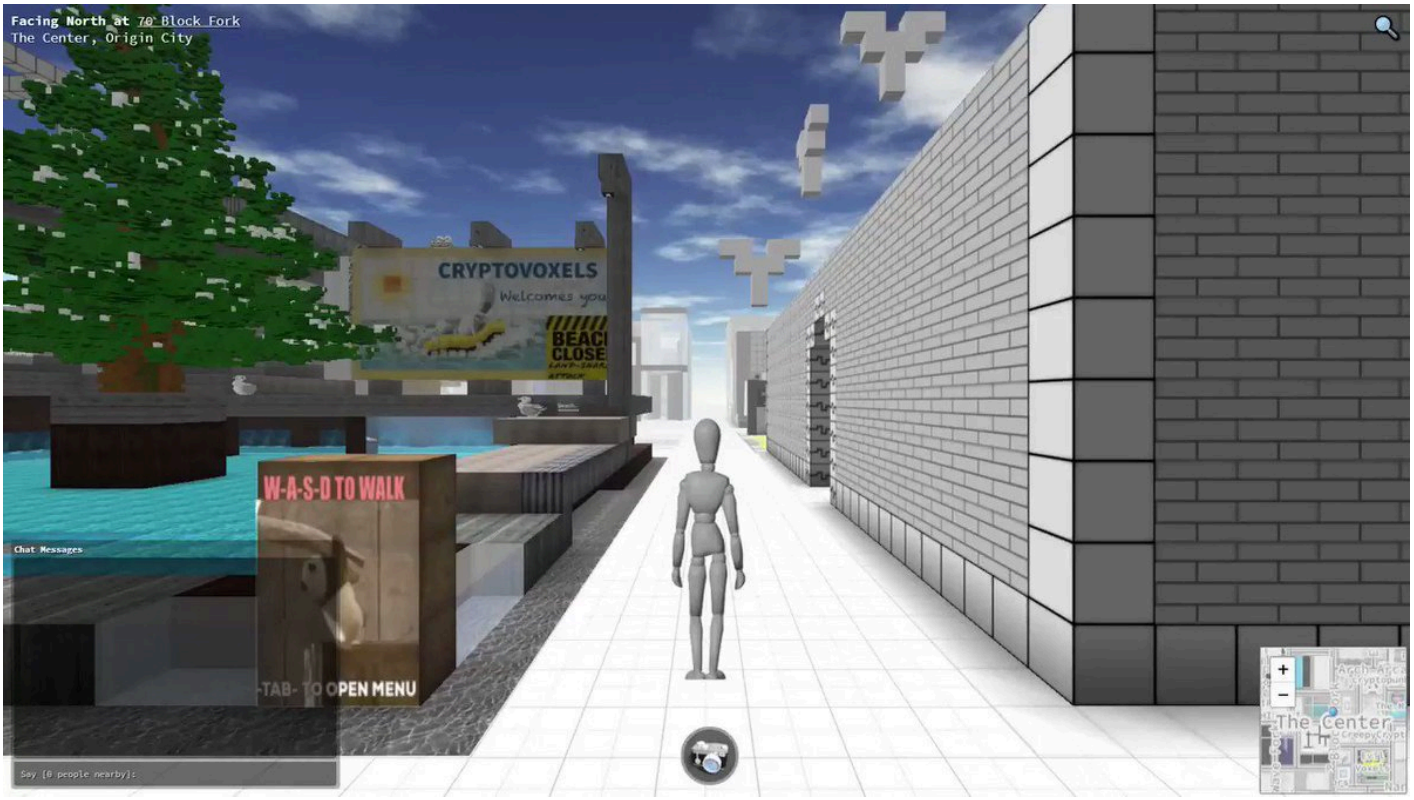
A long time ago Voxels implemented an avatar imposter system that replaced distant avatars with a sprite to improve rendering performance. However, the sprite had no resemblance to your costume so every low-performance mode avatar looked the same.

Cryptovoxels

@cryptovoxels

As part of scaling up, Marcus has made this 'sprite billboard system' that can render 100s of avatars on even low end machines.

Nearby avatars will be animated, distant ones will be a sprite like this. So awesome!



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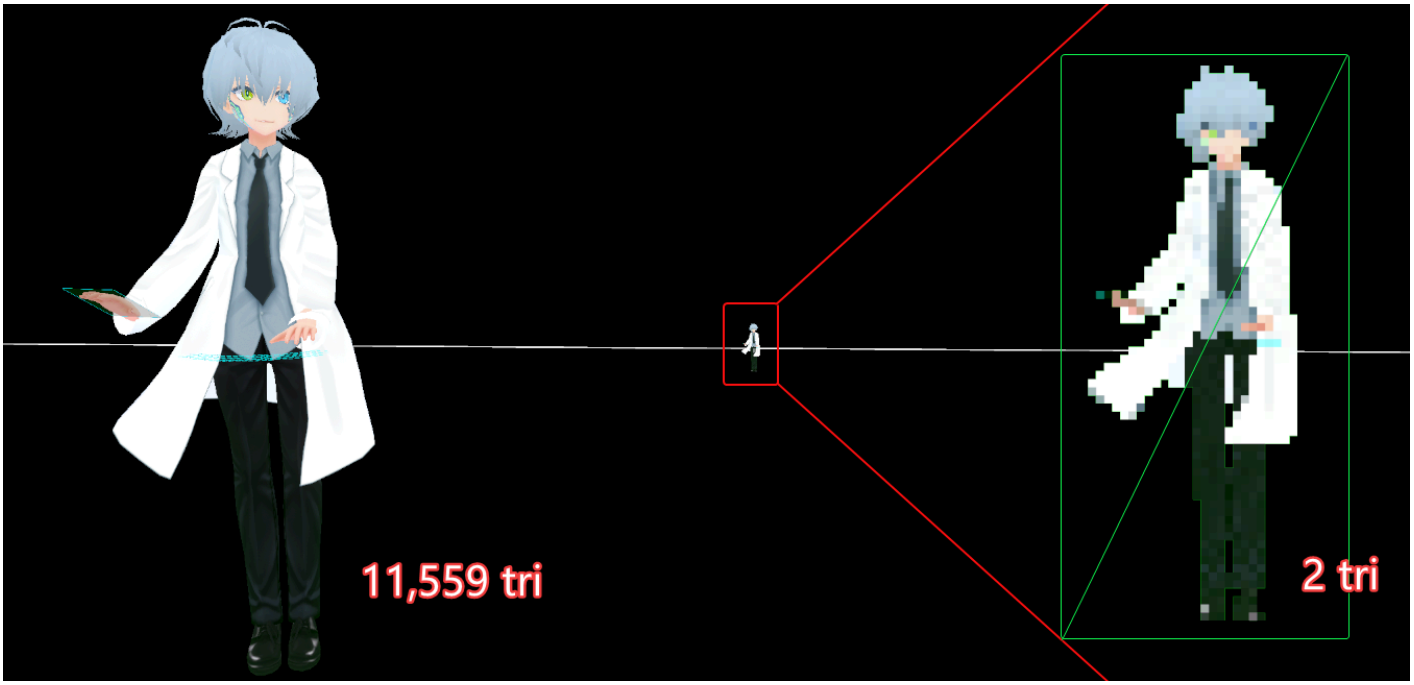
6:40 PM • Mar 25, 2020

This type of system is a relatively common technique for scaling the number of concurrent users in instances by the way. If you want to learn more see this dev log:

Avatar LOD - HackMD

Technical concepts to scale the number of simultaneous users per virtual worlds by an order of magnitude.

https://hackmd.io



However, usually these sprites are platform locked-in. What if you could generate a low poly version of your avatar and we worked towards standardization of avatar LOD? We implemented such a feature into Character Studio recently, here's the docs page.

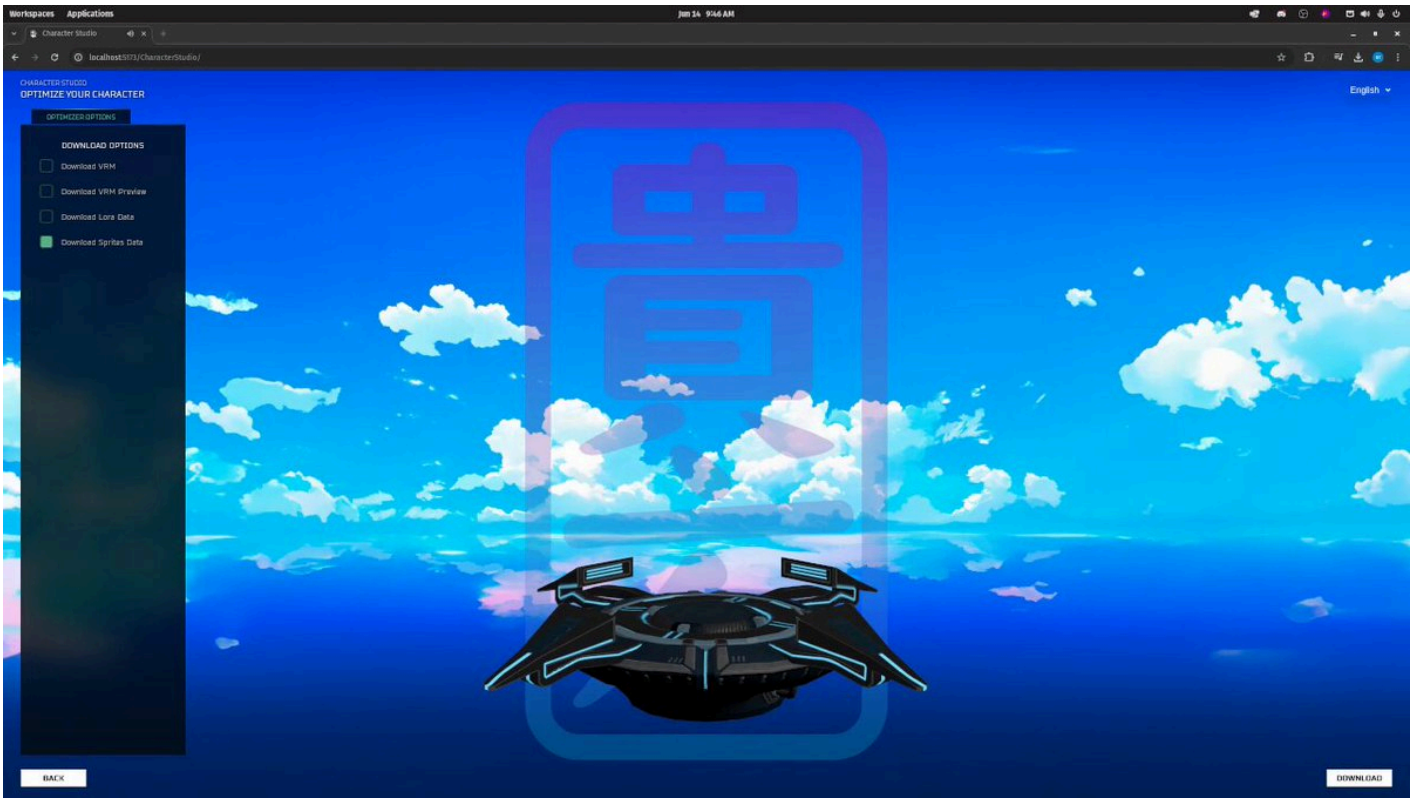


jin

@dankvr

WIP VRM -> sprite sheet pipeline via character studio. This time the code has been refactored to be more modular / inspired by Unix philosophy

Drag n drop VRM, custom animations supported, export zip with sprites #threejs #webgl



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10:25 AM • Jun 14, 2024

We processed all the Voxels VRMs through Character Studio in order to generate sprite sheets from them. Added a couple toe bones to the VRMs and re-exported them all so that the walk animation cycle was stable.



Every Voxel VRM will also get a sprite sheet version of their costume.

Openvoxels mentioned in Bankless

Context: following the announcement of the indie game studio VIPE shutting down, there was a Bankless post about building a sustainability and highlighted Openvoxels as one of the projects to check out.

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Openvoxels mentioned

"The group works to preserve Voxels artifacts and cultures, and it develops case study builds and new collaborative functionalities around Voxels assets."

Hopefully our case studies prove useful beyond Voxels someday. Overall a good and sobering read 

Bankless

@BanklessHQ



Building a Sustainable Metaverse

Lessons from the Virtual Land Collapse

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This week @polygonalmind, the team behind @vipeio, @cryptoavatarsio, and countless other virtual experiences, announced they were closing up shop because of



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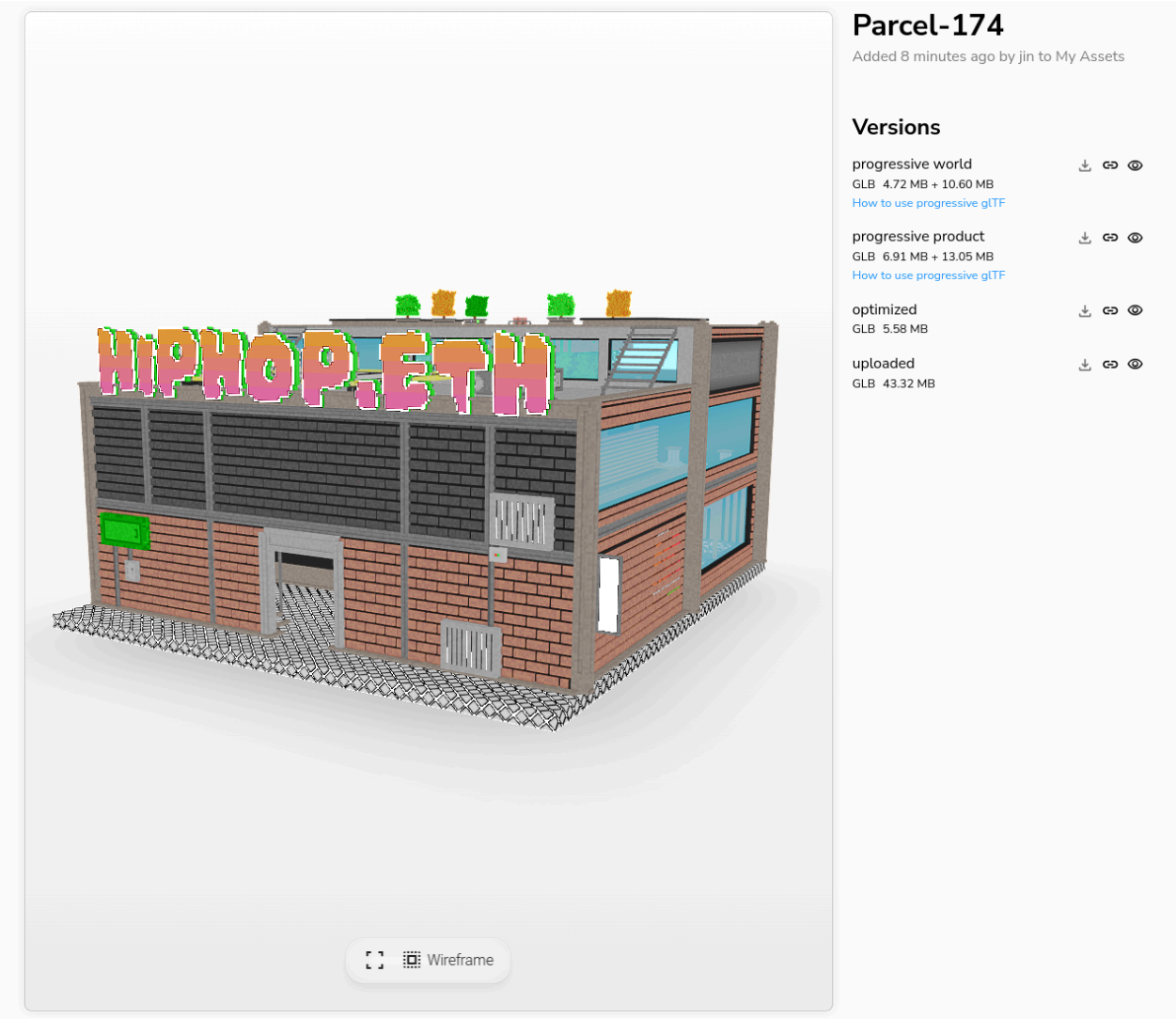
6:15 PM • Aug 4, 2024

Parcel Optimization Testing

Tested out needle cloud and gltf-transform on how well they can optimize a glTF export of a Voxels parcel. Below is parcel 174 which is a sample that Ben released last year when working on glTF export.

- Original size: 43.32 MB
- Needle Cloud Optimized: 5.58 MB

- gltf-transform optimize command: 7.45 MB



https://cloud.needle.tools/

Voxels Passports

Earlier in June we had an idea to generate 3D passports to go with the VRMs since most marketplaces don’t have a great VRM viewer yet. The passport would have all the information about the avatar contained within it. Well later we decided to work on it.

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Idea pitch for @cryptovoxels avatars:
- costume data embedded in VRM + NFT metadata
- NFT art styled like a passport (maybe even 3d)
- vrm_url in metadata
- link to sprite sheet in metadata
- maybe @MSquared_io support later
- airdropped to openvoxels supporters

Voxels Passports
Openvoxels.eth

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VRMs of current openvoxels supporters are ready, can request early access if you donated to the juicebox juicebox.money/@openvoxels

Thinking about doing a NFT collection for all the Voxel avatars, need to brainstorm more on how it can work



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10:36 PM • Jun 21, 2024

7OROY created an animated 3D passport in multiple different colors. We are thinking information about the owner of the costume like the ETH address + costume ID can fill some of the forms, and have information about currently equipped wearables also like a RPG inventory system. We generated a portrait picture of each VRM to replace the placeholder picture as well.

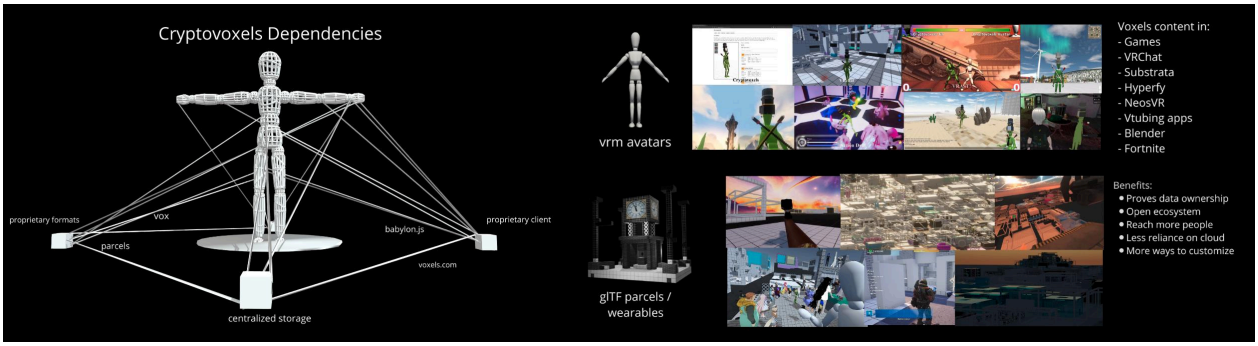


Preview of Voxels Passports, coming soon

September

Published a new post on 'transition thinking' and how niche projects like Openvoxels can be an agent for change:
https://mirror.xyz/m3org.eth/Mo9plGOkHyUSMObl3GPWpiwdX_XbJmQxdAB9kwnoZFA

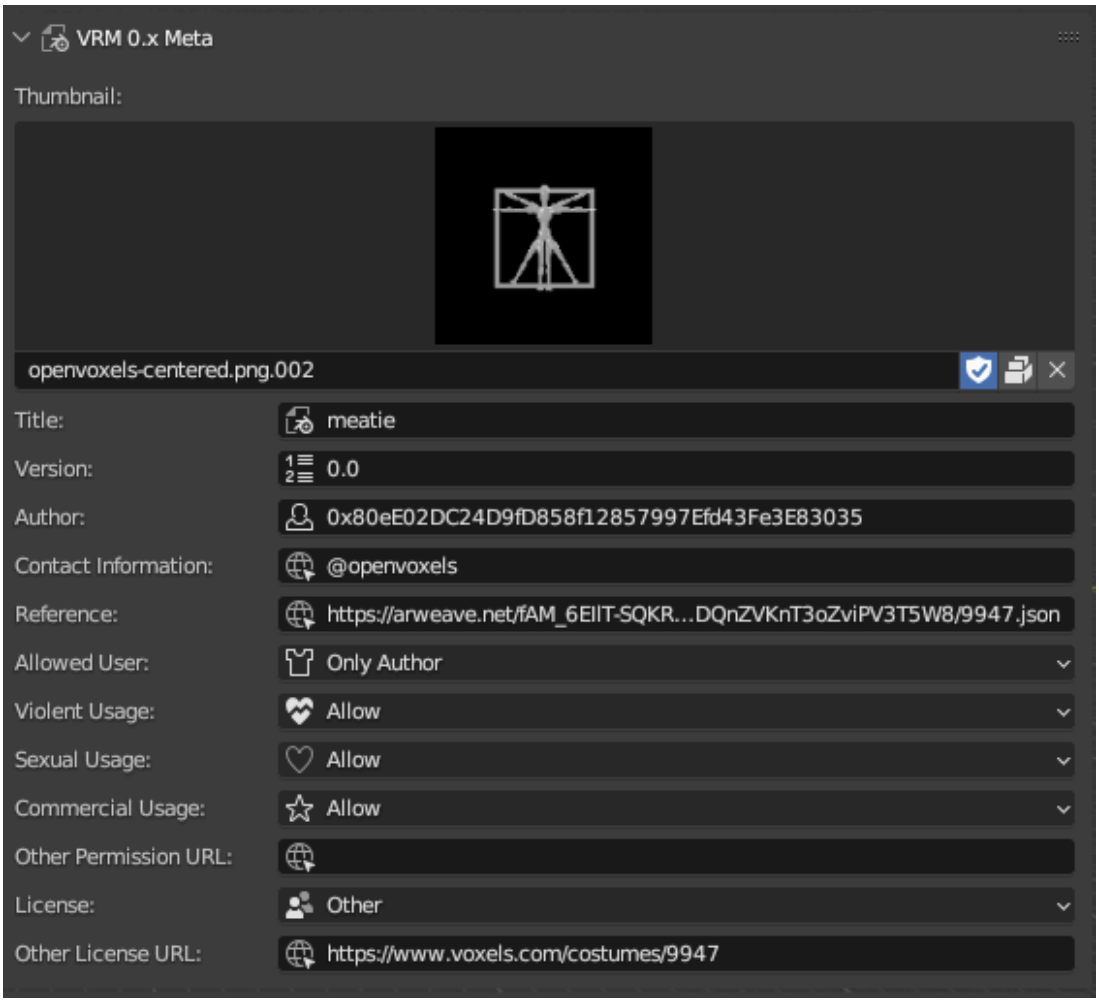
> Openvoxels aims to serve as a showcase for the potential benefits and challenges of granting users more ownership over our in-world items, a radical concept in the gaming industry. So far we've been influential in wider adoption of glTF and VRM, popular open file formats for content and avatars to move between immersive virtual world platforms.



Optimized Voxels VRMs + Metadata

We fixed a few bugs in Character Studio in order to optimize the Voxel VRM avatars to be 1 draw call each and hit a file size that can allow users to upload to all the major VRM supported web platforms including Oncyber which has a pretty strict upload limit.

All of the costumes were uploaded to arweave and get included in the reference field for the VRM metadata. The costumes contain information about the wearables such as the contract address + ID and the bone + transform data for where it’s attached to.



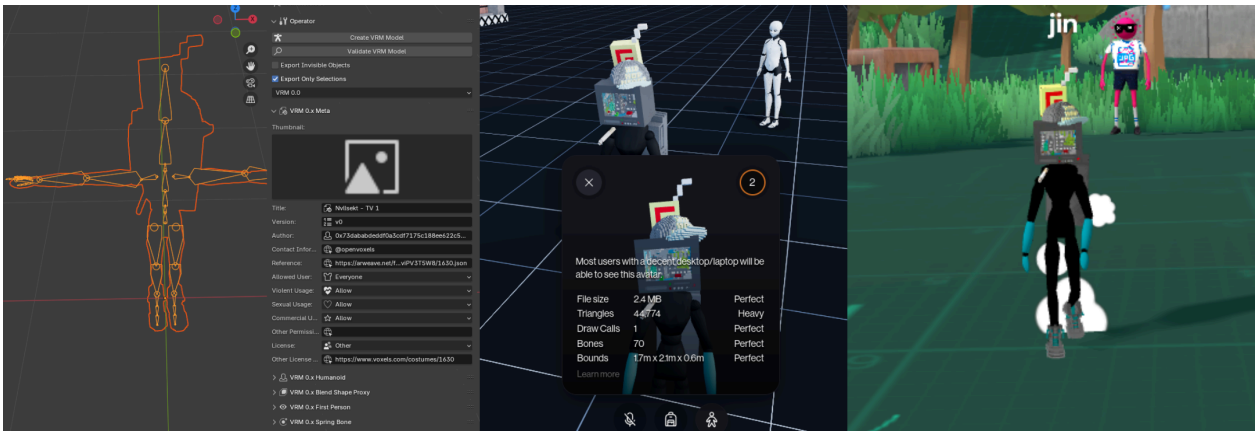
Example metadata that comes with every VRM

This may seem like a small detail but it's one I think is pretty important in terms of setting a standard. The optimization tools we used were stripping out metadata that we encoded into the VRMs, so we took a day to fix it in order to preserve that information throughout the whole cycle. Digital preservation / attribution is a pillar of the cryptoart scene, is it not?

VRM + KTX2 / Draco

There will be 2 versions of the 3D Voxel avatars:

- Original (LOD0): Separate meshes + materials per wearable
- Optmized (LOD1): Join meshes + materials, apply draco + KTX2 for smaller file size



On the left: VRM metadata preserved in optimized version, tested in Hyperfy / Oncyber also

All the avatars have been baked to 1 draw call each and uploaded to arweave. These optimizations allow platforms to render many more of them in a scene with better performance, as seen here in this stress test using an engine built on threejs.



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Stress testing 150 Voxels VRMs with [@AshConnell](#)



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Optimizing all Voxel VRMs, each will be 1 draw call
Texture atlas + KTX2 + Draco, ~60% smaller file size

Planning to provide original full size for customization + optimized versions for web in the NFT metadata. Gotta get platforms to somehow prioritize the optimized ones though

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9:27 AM • Sep 5, 2024

Project TBA

M3 is starting to explore the beginning of a protocol / virtual world builder inspired by The Street from Snow Crash. If you've read this far you deserve to know some details:

- Voxel editor for world building accessibility

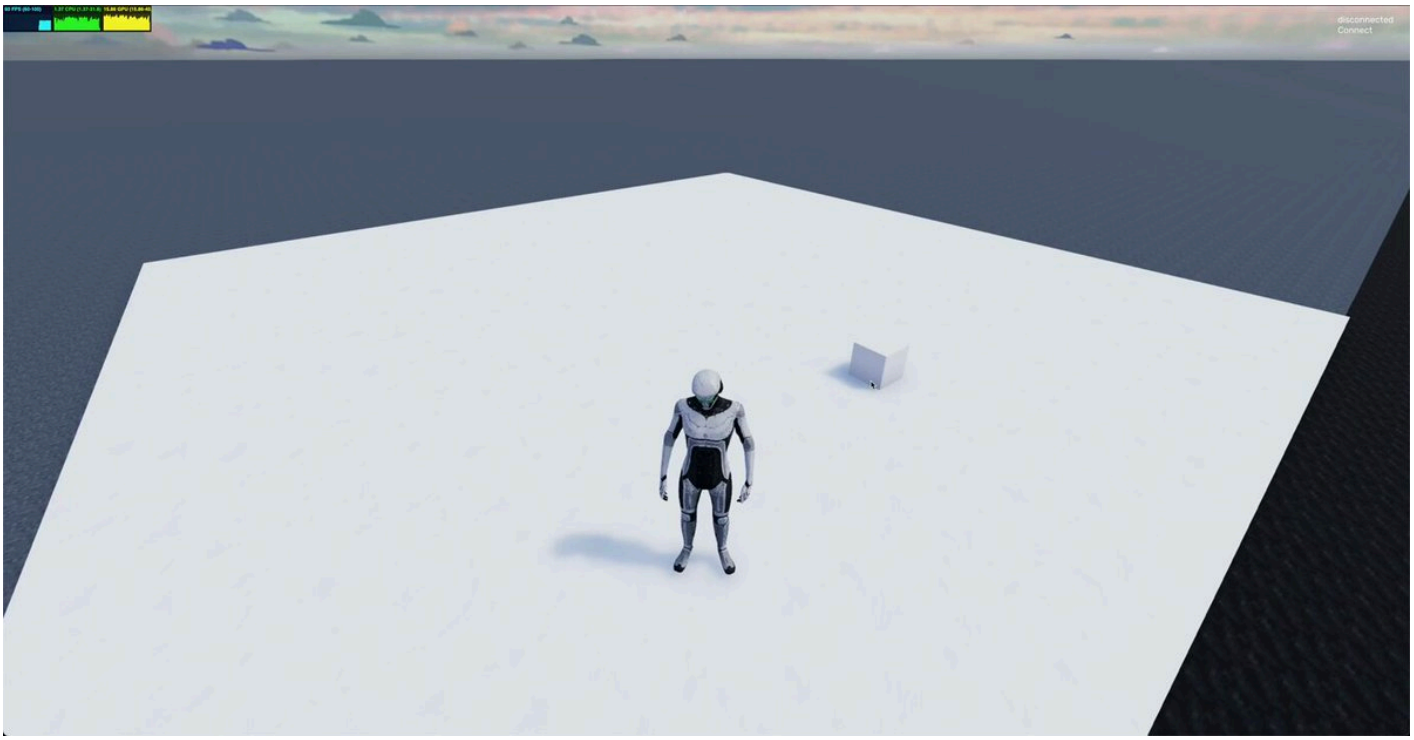
- VRM avatars for self expression
- glTF export day 1 (and regular world snapshots)



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the start of a basic voxel editor





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1:01 PM • Sep 6, 2024

Not planning to sell anything anytime soon. We aim to dogfood internally first (if you want to join M3 join the server and git gud) and then gradually open it up. That's all we'll say for now.

Big shoutout to all of our supporters, including a few new ones that recently donated to Openvoxels juicebox: Thank you octobar, curatorofcool, and DAPPCRAFT!!

Juicebox | Fund Your Thing

Juicebox is the programmable funding platform for crypto and web3. Fund, operate, and scale your project transparently. Community DAO owned, on Ethereum.

https://juicebox.money

